

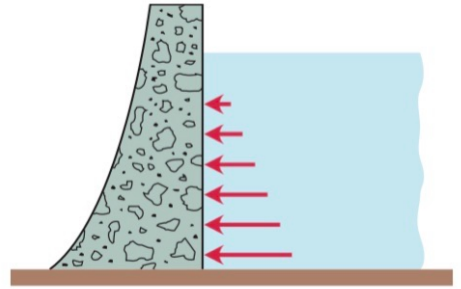
Lec 8-2, Wed, Oct 29.

- Fluid force (6.8)
- Inverse differentiation (7.1)
- Natural logarithmic function (7.2)
 - Basic properties
 - Algebraic properties
 - Graph and range
 - $\ln(|f(x)|)$

Fluid forces

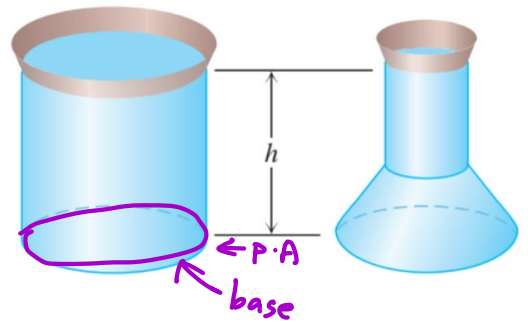
- The same object will generally experience bigger force if it is put in a deeper position in the same fluid.

- For standing-still fluid, the pressure p at depth h is $p = w \cdot h$, where w is the weight-density of the fluid [N/m^3 , say].



- For a container with a horizontal base, the total force applied by the fluid to base is

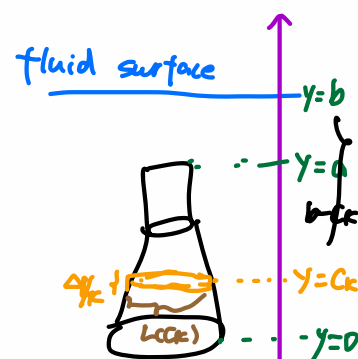
$$F = p \cdot A = w \cdot h \cdot A.$$



- If a flat plate is submerged vertically, then the pressure against it will depend on the depth of the part of the plate.
- To calculate the fluid force against it, we may use integrals.
- "Cut" the plate into horizontal slices S_k .

- Force exerted against the slice S_k is

$$\begin{aligned} F_k &= p_k \cdot A_k = w \cdot h_k \cdot A_k \\ &= w \cdot (b - c_k) \cdot L(c_k) \cdot \Delta y_k \end{aligned}$$



• Force exerted against the plate is

$$F = \lim_{n \rightarrow \infty} \sum_{k=1}^n w \cdot (b - c_k) \cdot L(c_k) \cdot \Delta y_k$$

$$= \int_0^a w (b - x) \cdot L(x) dx.$$

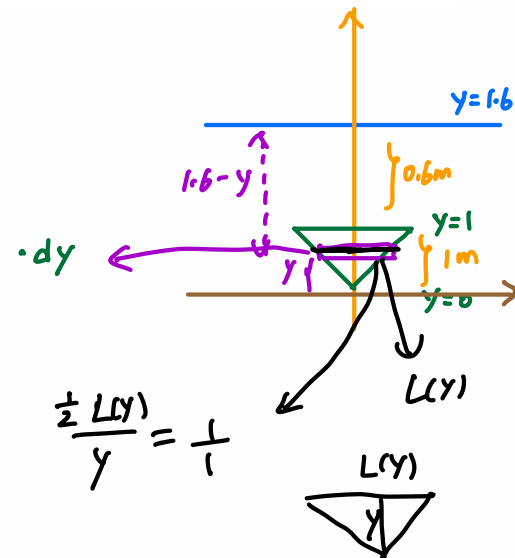
EXAMPLE 6.5.6 A flat isosceles right-triangular plate with base 2 m and height 1 m is submerged vertically, base up, 0.6 m below the surface of a swimming pool. Find the force exerted by the water against one side of the plate.

Sol': $F = \int_0^1 w (1.6 - y) \cdot L(y) dy$

$$= \int_0^1 w (1.6 - y) \cdot y dy$$

$$= w \left(1.6y^2 - \frac{2}{3}y^3 \right) \Big|_0^1$$

$$= 2w \left(0.8 - \frac{1}{3} \right),$$



where $w = 1000 \text{ kg/m}^3 \cdot 9.8 \text{ m/s}^2 = 9800 \text{ N/m}^3$ for water. $\square$

Summary.

① Riemann sum and definite integral

f is integrable on $[a, b]$ if and only if all the Riemann sums have a limit

$$\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k = J.$$

Then we denote this J by $\int_a^b f(x) dx$.

② Specifically, equal partition: $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \frac{b-a}{n} = \int_a^b f(x) dx$.

③ f is continuous or at most jumps finitely $\Rightarrow f$ is integrable.

f is unbounded on $[a, b]$ $\Rightarrow f$ is not integrable on $[a, b]$.

④ integration rules : linearity, additivity, max-min inequality ($\Rightarrow$ MVT)

$$\frac{1}{b-a} \int_a^b f(x) dx = f(c)$$

⑤ FTC 1. $\frac{d}{dx} F(x) = \frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$.

⑥ FTC 2. $\int_a^b f(t) dt = F(b) - F(a)$.

Compute the antiderivative!

[provide us a weapon to compute definite integral instead of limit of Riemann sums].

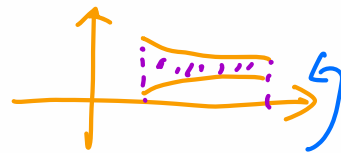
⑦ areas between curves

⑧ Substitution method: $\int_a^b f(g(x)) g'(x) dx = \int_a^b f(g(x)) d(g(x)) \stackrel{u=g(x)}{=} \int_{g(a)}^{g(b)} f(u) du$.

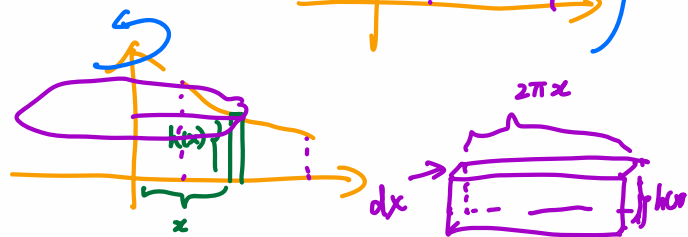
⑨ volume [cross-sections, cylindrical shells], arc length, $= F(g(x)) \Big|_a^b$

Area of surfaces of revolution, work, fluid forces.

$$\cdot V = \int_a^b A(x) dx = \int_a^b \pi [R(x)^2 - r(x)^2] dx$$



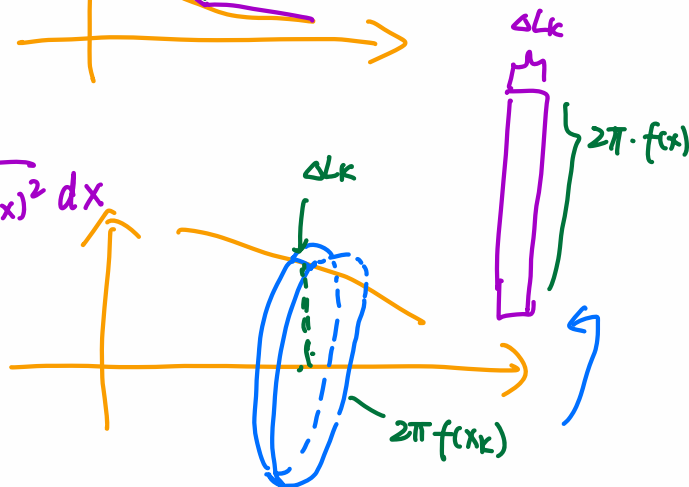
$$\cdot V = \int_a^b 2\pi x \cdot h(x) \cdot dx$$



$$\cdot L = \int_a^b \sqrt{1 + f'(x)^2} dx$$



$$\cdot S = \int_a^b 2\pi f(x) \cdot dL = \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} dx$$



$$\cdot W = \int_a^b F(x) dx$$

$$\cdot F = \int_0^a \underbrace{w \cdot (b-x)}_P \cdot \underbrace{L(x)}_A \cdot dx$$

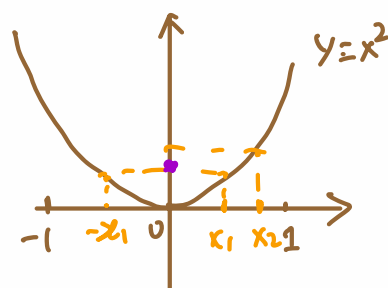
Chap 7.

Inverse Functions and Differentiation.

Def. A function f with domain D is one-to-one (or injective) if $f(x_1) \neq f(x_2)$ for all $x_1, x_2 \in D$ with $x_1 \neq x_2$.

e.g. $f(x) = x^2$ is one-to-one on $[0, 1]$,
but not on $[-1, 1]$.

$$[f(0.5) = f(-0.5) \text{ but } 0.5 \neq -0.5].$$



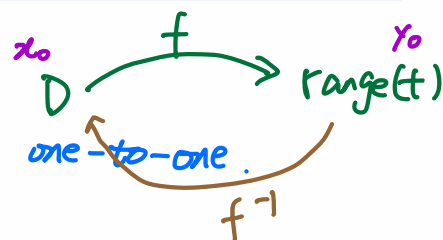
Def. Let $f : D \rightarrow \text{range}(f)$ be one-to-one. The inverse function of f is the function $f^{-1} : \text{range}(f) \rightarrow D$ defined by

$$f^{-1}(y_0) = x_0 \quad \text{where } f(x_0) = y_0.$$

Rmks. Inverse of f is only defined when f is one-to-one.

The range of f^{-1} is $\text{domain}(f) = D$.

By definition, $\forall x_0 \in D$, $\underline{f^{-1}(f(x_0)) = x_0}$.



$\forall y_0 \in \text{range}(f), \underline{f(f^{-1}(y_0))} = y_0$. So $f^{-1} \circ f$ and $f \circ f^{-1}$ are both identity functions

[a function that maps an element back to itself].

• $f: [0, 2] \rightarrow [0, 4]$. $f(x) = x^2 \stackrel{y}{=} \Rightarrow x = \sqrt{y}$ has inverse

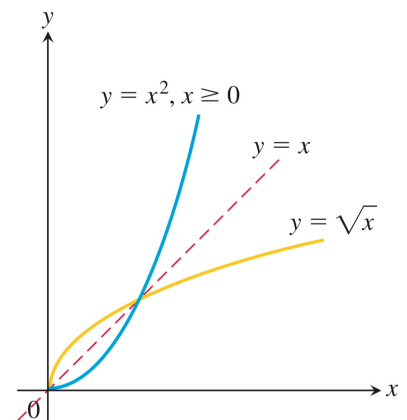
$$f^{-1}: [0, 4] \rightarrow [0, 2], \quad f^{-1}(y) = \sqrt{y}.$$

• Note that x and y are dummy variables, in the sense that

$f^{-1}(x) = \sqrt{x}$ and $f^{-1}(y) = \sqrt{y}$ represent the same function rule.

• If f is monotonic on D , then f is one-to-one on D ,

So inverse $f^{-1}: \text{range}(f) \rightarrow D$ always exists.



Fact (without proof)

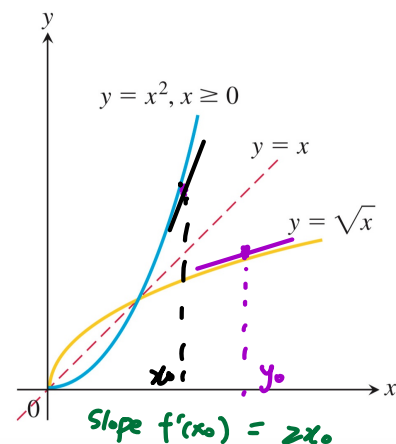
If $f: I \rightarrow \text{range}(f)$ is continuous and f^{-1} exists, then f^{-1} is also continuous.

Thm. [Inverse differentiation rule].

Let $f: I \rightarrow \text{range}(f)$ be one-to-one, where I is an interval. If f is differentiable and f' is never zero on I , then f^{-1} is differentiable and

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)} = \frac{1}{f'(f^{-1}(y_0))}, \quad \forall y_0 \in \text{range}(f),$$

where $x_0 = f^{-1}(y_0)$ (or $f(x_0) = y_0$).



$$\text{LHS} = (f^{-1})'(y_0) = \frac{1}{2} (y_0)^{-\frac{1}{2}} = \frac{1}{2\sqrt{y_0}}$$

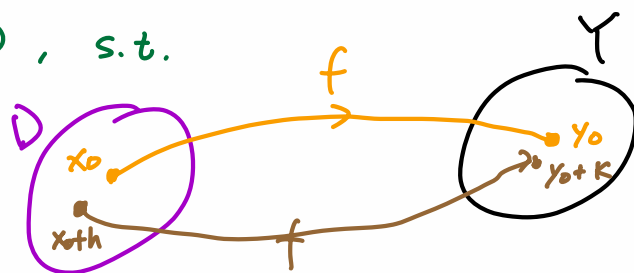
$$\text{RHS} = \frac{1}{f'(f^{-1}(y_0))} = \frac{1}{2 \cdot f^{-1}(y_0)} = \frac{1}{2\sqrt{y_0}}$$

Pf [optional].

Let $Y = \text{range}(f)$. Fix any interior point y_0 in Y .

For any $y_0 + k$, $\exists$ a unique $x_0 + h$ in D , s.t.

$$f(x_0 + h) = y_0 + k.$$



$$\lim_{k \rightarrow 0} \frac{f^{-1}(y_0 + k) - f^{-1}(y_0)}{k} = \lim_{k \rightarrow 0} \frac{f^{-1}(f(x_0 + h)) - f^{-1}(f(x_0))}{k}$$

$$= \lim_{k \rightarrow 0} \frac{x_0 + h - x_0}{k}$$

$$= \lim_{k \rightarrow 0} \frac{h}{y_0 + k - y_0}$$

$$= \lim_{k \rightarrow 0} \frac{h}{f(x_0 + h) - f(x_0)} \quad \textcircled{1}$$

- View h as a function of k .
- Since f is continuous, so is f^{-1} (by the fact above).
- Hence $h \rightarrow 0$ as $k \rightarrow 0$. [To see this,

$$h = x_0 + k - x_0$$

$$= f^{-1}(y_0 + k) - f^{-1}(y_0) \rightarrow 0 \quad \text{as } k \rightarrow 0,$$

$$\text{since } \lim_{k \rightarrow 0} f^{-1}(y_0 + k) = f^{-1}(y_0) \quad \text{.] .}$$

· By $\textcircled{1}$,

$$\lim_{k \rightarrow 0} \frac{f^{-1}(y_0 + k) - f^{-1}(y_0)}{k} = \lim_{h \rightarrow 0} \frac{h}{f(x_0 + h) - f(x_0)}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\frac{f(x_0 + h) - f(x_0)}{h}}$$

$$= \frac{1}{f'(x_0)}$$

[we use $f'(x_0) \neq 0$]

$$\therefore (f^{-1})'(y_0) \text{ exists and } = \frac{1}{f'(x_0)} .$$

• Proof is similar if y_0 is an endpoint of γ

□.

[use one-sided derivative].

e.g. Consider $f(x) = x^3 - 2$ defined on $(0, \infty)$.

(a) Find f^{-1} and its domain.

(b) Find $(f^{-1})'$

Sol: (a) Let $f(x) = x^3 - 2 = y \Rightarrow x = (y+2)^{\frac{1}{3}}$.

$\therefore f^{-1}(y) = (y+2)^{\frac{1}{3}}$, whose domain is $\text{Range}(f) = (-2, \infty)$.

(b) Method I: take the derivative of f^{-1} directly:

$$(f^{-1})'(y) = \frac{1}{3}(y+2)^{-\frac{2}{3}}, \quad y \in (-2, \infty).$$

Method II: Inverse differentiation.

$$(f^{-1})'(y) \stackrel{\text{Thm}}{=} \frac{1}{f'(x)} \Big|_{x=f^{-1}(y)}$$

$$(\because f(x) = x^3 - 2)$$

$$= \frac{1}{f'(f^{-1}(y))}$$

$$= \frac{1}{3(f^{-1}(y))^2}$$

$$= \frac{1}{3\left((y+2)^{\frac{1}{3}}\right)^2} = \frac{1}{3} (y+2)^{-\frac{2}{3}}, \quad y \in (-2, \infty). \quad \square$$

Natural logarithmic function

Def: (Saint-Vincent, 1649)

The natural logarithmic function is the function $\ln$ defined on $(0, \infty)$

by $\ln(x) \triangleq \int_1^x \frac{1}{t} dt$, $x \in (0, \infty)$

• High school: power function $10^x, 2^x \rightarrow \log$ func'

• College: log function $\ln(x) \rightarrow$ power

• Hence, $\ln(x)$ is the signed area under the curve $y = \frac{1}{t}$ from

$t=1$ to $t=x$, $x > 0$.

• In other textbooks, the natural logarithmic function is also written as $\log(x)$.

1. Basic properties

$$\textcircled{1} \ln 1 = \int_1^1 \frac{1}{t} dt = 0$$

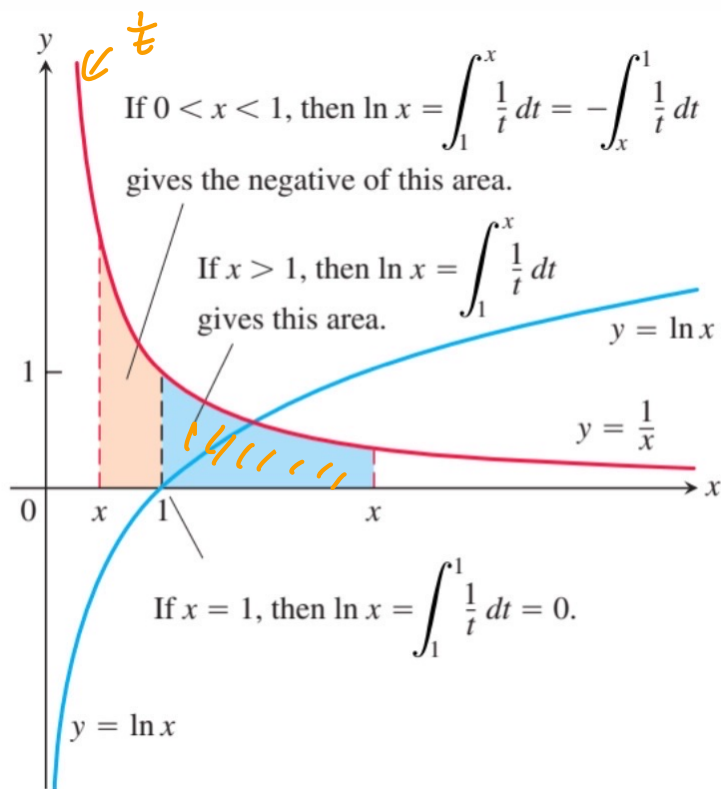
$$\textcircled{2} \ln' x = \frac{d}{dx} \left[\int_1^x \frac{1}{t} dt \right]$$

$$\stackrel{\text{FTC 1}}{=} \frac{1}{x}, \quad x > 0.$$

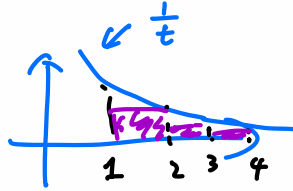
— Hence, $\ln$ is differentiable on $(0, \infty)$ and also continuous on $(0, \infty)$.

$$\textcircled{3} \because \ln'(x) = \frac{1}{x} > 0, \quad \forall x \in (0, \infty)$$

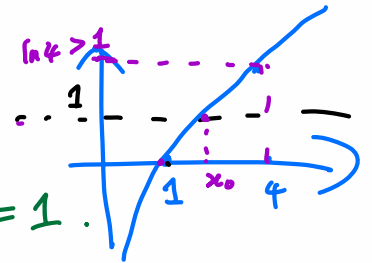
$\therefore \ln$ is increasing on $(0, \infty)$.



④ $\therefore \frac{1}{t}$ is decreasing of t



$$\begin{aligned}\therefore \ln 4 &= \int_1^4 \frac{1}{t} dt = \int_1^2 \frac{1}{t} dt + \int_2^3 \frac{1}{t} dt + \int_3^4 \frac{1}{t} dt \\ &\geq \frac{1}{2} \cdot 1 + \frac{1}{3} \cdot 1 + \frac{1}{4} \cdot 1 \\ &= \frac{6+3+2}{12} > 1.\end{aligned}$$



By IVT, there exists a number $x_0 \in (1, 4)$ s.t. $\ln(x_0) = 1$.

By Rolle's thm, this number is unique on $(0, \infty)$.

Def. Define e to be the unique number in $(0, \infty)$ such that

$$\ln(e) = 1.$$

Historical Remarks about e

- Bernoulli, 1683, in the context of compound interest.
- Euler, 1731, in the context of logarithm. [The notation " e " credits to Euler.]